

Diskrete Mathematik – Woche 2

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<https://github.com/tschai-yim/eth-resources>

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Propositional Logic

Problem 1 ★

Evaluate the truth value of the formula $(A \wedge \neg A) \rightarrow B$ for any arbitrary truth assignment of B . Justify your answer algebraically using Lemma 2.1 and the useful equivalences.

Solution:

$$\begin{aligned}(A \wedge \neg A) \rightarrow B &\equiv \neg(A \wedge \neg A) \vee B && \text{(Implication equivalence and lifting)} \\ &\equiv (\neg A \vee \neg \neg A) \vee B && \text{(De Morgan's rules)} \\ &\equiv (\neg A \vee A) \vee B && \text{(Double negation)} \\ &\equiv \top \vee B && \text{(Tautology)} \\ &\equiv \top && \text{(Constants)}\end{aligned}$$

Problem 2 ★

Determine whether the following expressions are syntactically valid formulas or statements about formulas. If an expression is invalid in both contexts, explain why.

1. $(A \rightarrow B) \equiv (\neg A \vee B)$
2. $(A \equiv B) \vee C$
3. $\models (A \leftrightarrow A)$
4. $A \leftrightarrow (B \leftrightarrow C)$

Solution:

1. **Statement.** $A \rightarrow B$ and $\neg A \vee B$ are valid formulas. $\equiv$ is a relation that takes two formulas to create a mathematical statement.
2. **Invalid.** $A \equiv B$ creates a mathematical statement, but the logical operator $\vee$ requires two propositional formulas as inputs, not statements.
3. **Statement.** $A \leftrightarrow A$ is a formula. $\models$ is a relation creating a statement expressing that the formula is a tautology (Definition 2.8).
4. **Formula.** A , B , and C are propositional symbols combined validly using the $\leftrightarrow$ operator.

Problem 3 ★ ★

Assume that $F \models G$ holds true for two propositional formulas F and G . If G is known to be unsatisfiable ($\perp$), what must be true about the formula F ? Proof your statement.

Solution:

F must be unsatisfiable. Assume for contradiction that F is satisfiable. Then there exists at least one truth assignment where F evaluates to $\top$. By Definition 2.7, since $F \models G$, G must also evaluate to $\top$ for this same assignment. However, we are given that G is unsatisfiable (it evaluates to $\perp$ for all assignments), which is a contradiction. Therefore, no such assignment exists for F , meaning F is unsatisfiable.

Problem 4 ★

Prove or disprove the following claim: “For any formulas F and G , if the formula $F \rightarrow G$ is satisfiable, then $F \models G$.”

Solution:

The claim is false. By Lemma 2.3, $F \models G$ if and only if $F \rightarrow G$ is a tautology (true for all assignments), which is a much stronger condition than satisfiability. Counterexample: Let $F = A$ and $G = B$. The formula $A \rightarrow B$ is satisfiable (e.g., evaluating to $\top$ when $A = \perp, B = \top$). However, $A \models B$ is false because if $A = \top$ and $B = \perp$, the premise is true but the conclusion is false.

Problem 5 (skip for time) ★ ★

Using the equivalences from Lemma 2.1 (specifically the distributive and absorption laws), prove that $(A \vee B) \wedge (A \vee \neg B) \equiv A$.

Solution:

$$\begin{aligned} (A \vee B) \wedge (A \vee \neg B) &\equiv A \vee (B \wedge \neg B) && \text{(Second distributive law (backwards))} \\ &\equiv A \vee \perp && \text{(Contradiction)} \\ &\equiv A && \text{(Constants)} \end{aligned}$$

Problem 6 (skip for time) ★ ★

Find a formula equivalent to $\neg(A \rightarrow (B \wedge C))$ that only uses the operators $\wedge, \vee$, and $\neg$, and where the $\neg$ operator only applies directly to the propositional symbols (literals).

Solution:

$$\begin{aligned} \neg(A \rightarrow (B \wedge C)) &\equiv \neg(\neg A \vee (B \wedge C)) && \text{(Implication equivalence)} \\ &\equiv \neg(\neg A) \wedge \neg(B \wedge C) && \text{(De Morgan's rules)} \\ &\equiv A \wedge \neg(B \wedge C) && \text{(Double negation)} \\ &\equiv A \wedge (\neg B \vee \neg C) && \text{(De Morgan's rules)} \end{aligned}$$

Predicate Logic

Problem 7 ★ ★

The script explicitly defines the Universe U as a **non-empty** set. Consider the logical consequence: $\forall x P(x) \models \exists x P(x)$. Explain why this holds true under the definition provided, but would fail if U were allowed to be the empty set $\emptyset$.

Solution:

If U is non-empty, there is at least one element $a \in U$. If $\forall x P(x)$ is true, then $P(a)$ must be true. Since $P(a)$ is true, the statement $\exists x P(x)$ is satisfied. If $U = \emptyset$, the premise $\forall x P(x)$ is vacuously true because there are no elements that violate $P(x)$. However, $\exists x P(x)$ is false because there is no element in U to satisfy the predicate. A true premise yielding a false conclusion breaks the logical consequence.

Problem 8 ★ ★

Consider the formula $F = \forall x \exists y (y < x)$. Evaluate its truth value under the following interpretations:

1. $U = \mathbb{N}$ (natural numbers 0, 1, 2, ...) and $<$ is the standard “less than” predicate.
2. $U = \mathbb{Z}$ (integers) and $<$ is the standard “less than” predicate.

Solution:

1. **False over $\mathbb{N}$.** If we choose $x = 0$, there is no natural number y such that $y < 0$. The universal claim fails.
2. **True over $\mathbb{Z}$.** For every integer x , we can always pick $y = x - 1$, which is an integer strictly smaller than x .

Problem 9 ★

Rewrite the following formulas using a new variable y so that the quantifier scopes do not overlap, preserving the exact same logical meaning.

1. $\exists x (x < 0) \wedge \exists x (x > 0)$
2. $\forall x (P(x) \rightarrow \exists x (Q(x)))$

Solution:

According to the script, variables with the same name in different scopes are independent.

1. $\exists x (x < 0) \wedge \exists y (y > 0)$
2. $\forall x (P(x) \rightarrow \exists y (Q(y)))$

Problem 10 ★ ★

Let U be the set of all animals. Let $C(x)$ mean “ x is a cat”, $M(x)$ mean “ x is a mammal”, and $T(x)$ mean “ x has a tail”. Translate the following statement into a single predicate logic formula: “All cats are mammals, but not all cats have tails.”

Solution:

“All cats are mammals” translates to a universal implication: $\forall x(C(x) \rightarrow M(x))$. “Not all cats have tails” translates to $\neg \forall x(C(x) \rightarrow T(x))$, which is logically equivalent to $\exists x(C(x) \wedge \neg T(x))$. Combining both parts with a logical AND yields: $\forall x(C(x) \rightarrow M(x)) \wedge \exists x(C(x) \wedge \neg T(x))$

Problem 11 (skip for time) ★ ★ ★

Let $U = \mathbb{N}$. Let $E(x)$ be the predicate “ x is even”, and $P(x)$ be “ x is prime”. Using only the standard quantifiers ($\forall, \exists$) and equality ($=$), write a formula expressing: “There exists exactly one even prime number.”

Solution:

Unique existence construction: $\exists x(F(x) \wedge \forall y(F(y) \rightarrow y = x))$. We substitute the combined property $(E(x) \wedge P(x))$ for $F(x)$: $\exists x((E(x) \wedge P(x)) \wedge \forall y((E(y) \wedge P(y)) \rightarrow y = x))$

Problem 12 ★ ★

Prove or disprove the following claims regarding logical consequence:

1. $\exists xP(x) \wedge \exists xQ(x) \models \exists x(P(x) \wedge Q(x))$
2. $\forall x(P(x) \vee Q(x)) \models \forall xP(x) \vee \forall xQ(x)$
3. $\forall xP(x) \wedge \forall xQ(x) \models \forall x(P(x) \wedge Q(x))$ (★ ★ ★)

Solution:

1. **False.** Let $U = \mathbb{N}$, let $P(x)$ mean “ x is even”, and $Q(x)$ mean “ x is odd”. The premise is true ($\top$) because there are even numbers and odd numbers. The conclusion is false ($\perp$) because no single number is both even and odd. Since $\top \not\models \perp$, the consequence fails.
2. **False.** Using the same interpretation, the premise is true (every number is either even or odd). The conclusion is false (it is false that all numbers are even, and false that all numbers are odd; $\perp \vee \perp \equiv \perp$).
3. **True.** Let I be an arbitrary interpretation with a universe U where the premise $\forall xP(x) \wedge \forall xQ(x)$ evaluates to $\top$. By definition of the $\wedge$ operator, both $\forall xP(x)$ and $\forall xQ(x)$ evaluate to $\top$. By definition of the $\forall$ quantifier, for any arbitrary element $a \in U$, $P(a) = \top$ and $Q(a) = \top$. Because both are true for a , the conjunction $P(a) \wedge Q(a) = \top$. Since a was chosen arbitrarily, $P(a) \wedge Q(a) = \top$ holds for all elements in U . Therefore, by definition of $\forall$, the conclusion $\forall x(P(x) \wedge Q(x))$ evaluates to $\top$ under I . Since this holds for any interpretation, the logical consequence is proven.

Problem 13 ★ ★

Find a formula equivalent to $\neg(\forall x \exists y \forall z (P(x, y) \rightarrow Q(y, z)))$ where the negation operator $\neg$ only applies directly to the predicates P and Q . Show your step-by-step equivalence derivations.

Solution:

$$\begin{aligned}
& \neg(\forall x \exists y \forall z (P(x, y) \rightarrow Q(y, z))) \\
& \equiv \exists x \neg(\exists y \forall z (P(x, y) \rightarrow Q(y, z))) \quad (\text{Negating } \forall) \\
& \equiv \exists x \forall y \neg(\forall z (P(x, y) \rightarrow Q(y, z))) \quad (\text{Negating } \exists) \\
& \equiv \exists x \forall y \exists z \neg(P(x, y) \rightarrow Q(y, z)) \quad (\text{Negating } \forall) \\
& \equiv \exists x \forall y \exists z \neg(\neg P(x, y) \vee Q(y, z)) \quad (\text{Implication equivalence}) \\
& \equiv \exists x \forall y \exists z (\neg \neg P(x, y) \wedge \neg Q(y, z)) \quad (\text{De Morgan's rules}) \\
& \equiv \exists x \forall y \exists z (P(x, y) \wedge \neg Q(y, z)) \quad (\text{Double negation})
\end{aligned}$$

Problem 14 (skip for time) ★

Let $U = \mathbb{Z}$ (integers) and $S(x, y)$ be the predicate $x + y = 0$. Determine the truth values of the statements $S_1 = \forall x \exists y S(x, y)$ and $S_2 = \exists y \forall x S(x, y)$.

Solution:

- S_1 is **True**. Because $\exists y$ comes after $\forall x$, y can depend on x . For any chosen x , we can pick $y = -x$ to satisfy $x + y = 0$.
- S_2 is **False**. Because $\exists y$ comes first, a single fixed y must work for every x . No single integer can act as the additive inverse for all integers simultaneously.